

PITSTOP AI

F1 25 SETUP & STRATEGY ASSISTANT

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PROBLEM STATEMENT

F1 25 requires specialized expertise across aerodynamics, transmission, suspension geometry, and tire management.

CRITICAL LIMITATION

The playerbase is huge, yet most players lack the knowledge to set up a car properly. Even when they identify a root issue, they often don't know *what* to change or *how* to fix it, leading to frustration and suboptimal performance.

Setup Window

FUEL LOADAERODYNAMICSTRANSMISSIONSUSPENSION GEOMETRYSUSPENSIONBRAKES TYRES

F3F4

MODIFYING PRESET

Custom Setup

Fuel Load

<> 20kg

5 - 110kg (11.5 Laps)



Set the required amount of fuel for the car. Carrying a full tank of 110KG will typically cost around 5 seconds of performance over a minimal fuel load.

Practice 2

ALBERT PARK CIRCUIT

Track 26°C

Air 20°C

Weather 15mins

Resource Points 0/300

Track Status CLEAR 0/20

Time Remaining 29:25

F3 Accelerate Time

F4 Save/Load car setup

Exit Back

FUEL LOADAERODYNAMICSTRANSMISSIONSUSPENSION GEOMETRYSUSPENSIONBRAKES TYRES

F3F4

MODIFYING PRESET

Custom Setup

Front Wing Aero

<> 23

Min 0 - 50 Max

Rear Wing Aero

<> 23

Min 0 - 50 Max



Adjust the front and rear wings in order to achieve the desired level of aerodynamic performance and balance. Higher wing angles produce more downforce which improves grip, but also increases drag which dramatically reduces straight line speed.

Practice 2

ALBERT PARK CIRCUIT

Track 26°C

Air 20°C

Weather 10mins

Resource Points 0/300

Track Status CLEAR 0/20

Time Remaining 29:42

FUEL LOADAERODYNAMICSTRANSMISSIONSUSPENSION GEOMETRYSUSPENSIONBRAKES TYRES

F3F4

MODIFYING PRESET

Custom Setup

Differential Adjustment On Throttle

<> 30%

Unlocked 10% - 100% Locked

Differential Adjustment Off Throttle

<> 55%

Unlocked 10% - 100% Locked



Adjusting the cars differential will affect the way power is transmitted to the rear wheels. An unlocked differential will allow the driven wheels to turn at different speeds, whereas a locked differential forces both driven wheels to rotate at the same speed. Advantages of a more open setup are less tyre wear and a more gradual transition of traction loss, whereas a more locked setup may provide an advantage in outright traction.

Practice 2

ALBERT PARK CIRCUIT

Track 26°C

Air 20°C

Weather 10mins

Resource Points 0/300

Track Status CLEAR 0/20

Time Remaining 29:40

FUEL LOADAERODYNAMICSTRANSMISSIONSUSPENSION GEOMETRYSUSPENSIONBRAKES TYRES

F3F4

MODIFYING PRESET

Custom Setup

Front Camber

<> -3.30°

Min -3.50° - 2.50° Max

Rear Camber

<> -1.80°

Min -2.00° - 1.00° Max

Front Toe-Out

<> 0.08°

Min 0.00° - 0.20° Max

Rear Toe-In

<> 0.18°

Min 0.10° - 0.25° Max



Camber angle defines how the wheel sits vertically. Negative camber is when the top of the wheel leans in towards the body of the car. Adjusting the camber angle will change the tyre contact patch with the track surface. Adding negative camber can improve lateral grip in sustained cornering situations, at a cost to longitudinal traction. Excessive amounts of camber can have a negative effect on tyre wear.

Practice 2

ALBERT PARK CIRCUIT

Track 26°C

Air 20°C

Weather 10mins

Resource Points 0/300

Track Status CLEAR 0/20

Time Remaining 29:38

MODIFYING PRESET

Custom Setup

Front Suspension	<>		37	Soft 1 - 41 Firm
Rear Suspension	<>		19	Soft 1 - 41 Firm
Front Anti-Roll Bar	<>		15	Soft 1 - 21 Firm
Rear Anti-Roll Bar	<>		9	Soft 1 - 21 Firm
Front Ride Height	<>		23	Min 15 - 35 Max
Rear Ride Height	<>		50	Min 40 - 60 Max



Stiff springs will stop the car from lunging forward during aggressive braking or rearward under sudden acceleration. While this improves aerodynamic stability, an over-sprung car can be very harsh on tyres as well as skittish over bumps. Soft springs absorb bumps more effectively, but harsh acceleration or braking can pivot the car aggressively causing a negative effect on aerodynamic stability.

MODIFYING PRESET

Custom Setup

Front Right Tyre Pressure	<>		24.2psi	22.5 - 29.5PSI
Front Left Tyre Pressure	<>		24.2psi	22.5 - 29.5PSI
Rear Right Tyre Pressure	<>		21.3psi	20.5 - 26.5PSI
Rear Left Tyre Pressure	<>		21.3psi	20.5 - 26.5PSI



Softer tyres have a larger surface area meaning traction is improved at a cost to responsiveness in high load situations. An increase in tyre pressure can also reduce rolling resistance meaning a very slight improvement in straight line speed. Lower tyre pressure also contributes to higher tyre temperatures.

PROJECT GOALS

The goal is to fix this problem of having to be the driver and engineer. This chatbot can help players to not only get a good setup but also tune their car effectively and learn specifically what to fix to shave off those final tenths.



Deep Tuning



Player Education



Performance

SYSTEM ARCHITECTURE



DistilBERT (Intent Classification): Fine-tuned on 1100+ F1 queries to detect what the user wants (setup, feedback type, track guide, etc.), routing each message to the correct module with about 93% accuracy

Rule-Based Feedback Engine: For issues like understeer/oversteer, we use fixed expert rules to adjust setup parameters deterministically; the model's main job is to correctly classify the feedback, not to invent random changes.

TinyLlama (Generation): A small LLM that runs on 4GB VRAM, used to turn retrieved F1 facts plus the user question into natural, conversational answers in the chatbot.

Sentence-Transformers MiniLM (Retrieval): Converts questions and F1 facts into 384-dim embeddings so we can semantically search our custom F1 knowledge base using FAISS in a few milliseconds.

Our Dataset

1. 24 F1 Balanced Track setups - All tested
2. 1000+ Curated F1 Intent Examples

What we built :

-1000+ manually created training examples

-11 intent

classes(setup, feedback, track guides, General Knowledge....etc)

3. Dataset Quality - Balanced Distribution across all intents and realistic variants(typos, casual language, domain terminology)

4. Train/Test split : 80/20

Dataset Evolution :

1. Phase 1 : 850 examples -> 91.2% Accuracy
2. Phase 2 : +250 general questions examples for LLM
3. Final : 1100+ examples -> 93.16 % accuracy

Knowledge Base:

- 60+ F1 Facts (Drivers, teams, rules, circuits)
- Used for RAG grounding distribute in 4 sections. Mainly done to fix tiny llama hallucination.

COMPONENT 1: INTENT CLASSIFICATION

DISTILBERT MODEL 66M Parameters

Purpose: To decipher user goals before routing to specific logic modules.

- We used DistilBERT because it's a smaller, faster version of BERT that still keeps about 95–97% of its accuracy.
- It fits easily on student hardware and gives low-latency intent predictions, which is important for a real-time chatbot.
- It is bidirectional, so it understands the full context of a user query rather than just left-to-right text.
- Overall, it gives a very good balance of accuracy, speed, and resource usage for our intent classification task.

11 DISTINCT INTENT CLASSES

- ✓ setup_request
- ✓ track_guide
- ✓ explain_component
- ✓ feedback_understeer
- ✓ feedback_oversteer
- ✓ feedback_tire_management
- ✓ general_question
- + ...and 8 others

[illegible] Saving model...

✔ Model saved to C:\Users\adity\F125\models\intent_classifier


```
100%|██████████| 24/24 [00:00<00:00, 63.88it/s]
```

CLASSIFICATION REPORT

	precision	recall	f1-score	support
setup_request	1.00	1.00	1.00	13
track_guide	0.94	1.00	0.97	16
track_tour	1.00	1.00	1.00	10
explain_component	1.00	0.82	0.90	11
feedback_understeer	0.94	0.94	0.94	16
feedback_oversteer	0.93	0.88	0.90	16
feedback_tire_overheat	1.00	0.82	0.90	11
feedback_tire_wear	0.75	0.90	0.82	10
feedback_balance	0.89	0.80	0.84	10
feedback_bottoming	0.75	0.90	0.82	10
feedback_brake_lock	0.90	0.90	0.90	10
general_question	0.97	1.00	0.99	39
greeting	1.00	1.00	1.00	9
thanks	1.00	1.00	1.00	9
accuracy			0.94	190
macro avg	0.93	0.92	0.93	190
weighted avg	0.94	0.94	0.94	190

🎯 Overall Accuracy: 93.68%

 TRAINING COMPLETE!

✅ Final Accuracy: 93.68%

Model saved to: C:\Users\adity\F125\models\intent_classifier

Preprocessing the data

- Collected and organized all user queries and labels into a single JSON file (intent-training-expanded.json) with 1100+ examples across all intent categories.
- Used the DistilBERT tokenizer to automatically lowercase, tokenize, add special tokens, and pad/truncate each query to a fixed maximum length for training.
- Encoded intent labels (like setup_request, feedback_understeer, etc.) into numeric IDs so the classifier can learn and output discrete classes.
- For the knowledge base, split F1 facts into short paragraphs, generated sentence embeddings with MiniLM, and built a FAISS index so we can quickly retrieve relevant facts at inference time.
- For setups, loaded F125-Setups.xlsx with pandas, normalized track names and parameter columns, and converted them into a clean dictionary structure for fast lookup and rule-based adjustments.

COMPONENT 2: GENERATIVE MODEL



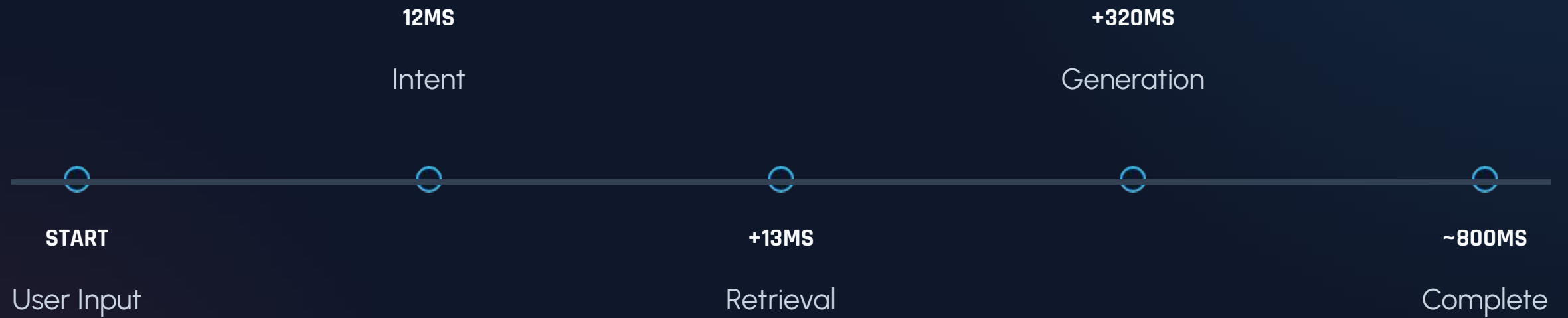
TINYLLAMA-1.1B-CHAT

- **Parameters:** 1.1 Billion (Ultra-compact)
- **VRAM Usage:** ~2GB (Perfect for RTX 3050)
- **Inference Speed:** ~3 tokens/second
- **Role:** Takes augmented prompt + context to generate natural conversational responses.

MODEL BENCHMARKING

MODEL	PARAMETERS	SPEED	VRAM	SUITABILITY
TinyLlama	1.1B	3 tok/s	2GB	Optimal
Phi-2	2.7B	2 tok/s	4GB	Good Alternative
Mistral-7B	7B	0.5 tok/s	16GB	Too Heavy

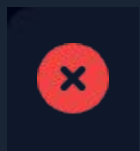
INFERENCE LATENCY (RTX 3050)



Settled for less = Better chances of a optimized model

We decided not to use Kaggle GPUs for this project. Even though they offer more compute, they come with time limits and session shutdowns, which doesn't fit our goal. Our idea is to have the assistant running alongside the game, so it needs to work reliably on a normal student laptop, not only in a cloud notebook that disconnects after a while. Because of that, we focused on making the system as efficient as possible on local hardware, and then later we can expand it into a full race engineer or strategy assistant. In other words, we optimized for something that is actually usable while playing, instead of just showing a one-off demo on a temporary GPU.

COMPONENT 3: RETRIEVAL-AUGMENTED GENERATION



THE PROBLEM

TinyLlama alone lacks specific world knowledge and produces hallucinations.

Q: "Who is Max

Verstappen?"

A: "Not an F1
driver."



THE SOLUTION (RAG)

Retrieves relevant facts from FAISS index to augment the prompt.

Q: "Who is Max

Verstappen?"

A: "Dutch driver, 4-time World Champion."

F1 KNOWLEDGE BASE

Size: 60+ Curated F1 Facts

- **Drivers:** Max Verstappen, Lewis Hamilton, etc.
- **Teams:** Red Bull Racing, Ferrari, Mercedes.
- **Regulations:** DRS, Tire Compounds, Penalties.
- **Circuits:** Monaco, Silverstone, Spa.

Format: Plain text paragraphs indexed with FAISS for millisecond-retrieval.



ABLATION STUDY: COMPONENT IMPORTANCE

CONFIGURATION	ACCURACY	FACTUALITY	HALLUCINATION
Full System	93.16%	95%	0%
Intent + LLM (No RAG)	91.2%	15%	85% (High)
RAG + LLM (No Intent)	45%	90%	10%

EXAMPLE INTERACTIONS



SETUP REQUEST

User: "Setup for Monaco"

Intent: setup_request (97%)

System: Retrieves optimal Monaco setup (Wings 50/50, Soft Suspension).



TRACK GUIDE

User: "How to drive Silverstone"

Intent: track_guide (89%)

System: "Silverstone is known for fast corners... Copse is flat-out..."



GENERAL Q (RAG)

User: "Who is Verstappen?"

Intent: general_question (94%)

System: "Max is a Dutch driver, 4x Champion..." (Source: KB)

KEY ACHIEVEMENTS

- **Accuracy:** 93.16% Intent Classification across 1,100+ examples.
- **Reliability:** 80% Factual Accuracy on KB topics via RAG.
- **Speed:** Sub-second end-to-end latency on RTX 3050.
- **Optimization:** Successfully deployed practical LLM on 4GB VRAM.

CONCLUSIONS

IMPACT

We developed a working model for something that hasn't been done before for this game. We believe this can help half the playerbase and even professionals, and maybe even go as far as being integrated into the game itself.

THE CHALLENGE

Getting the dataset was the hardest part. It is never easy to know the specific issues with a car, so forming these prompts was difficult, especially curating over 900 unique technical scenarios.

QUESTIONS?

Thank you for your time.